OLG and Pension Policies

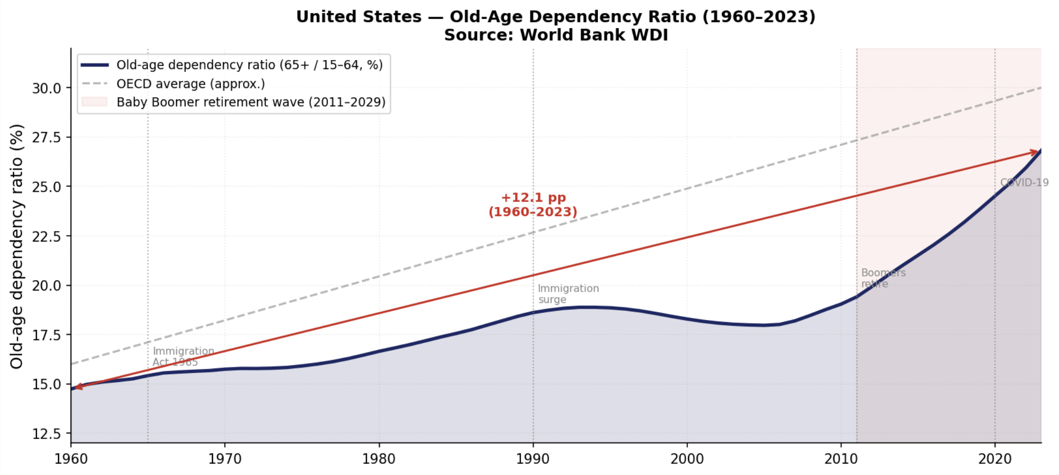
Contents

Pensions: Some data

Public Pensions

- Public pay-as-you-go pensions were introduced in Germany in 1889 and spread across Europe and the US in the early twentieth century.
- Since then, longevity has increased substantially, so the fiscal burden of pension systems has risen.
- This immediately raises three central questions for OLG analysis:
  - Are public pensions sustainable?
  - How should modern pension systems be reformed?
  - What is the welfare-maximizing pension level?

U.S. Old-Age Dependency Ratio and its Determinants



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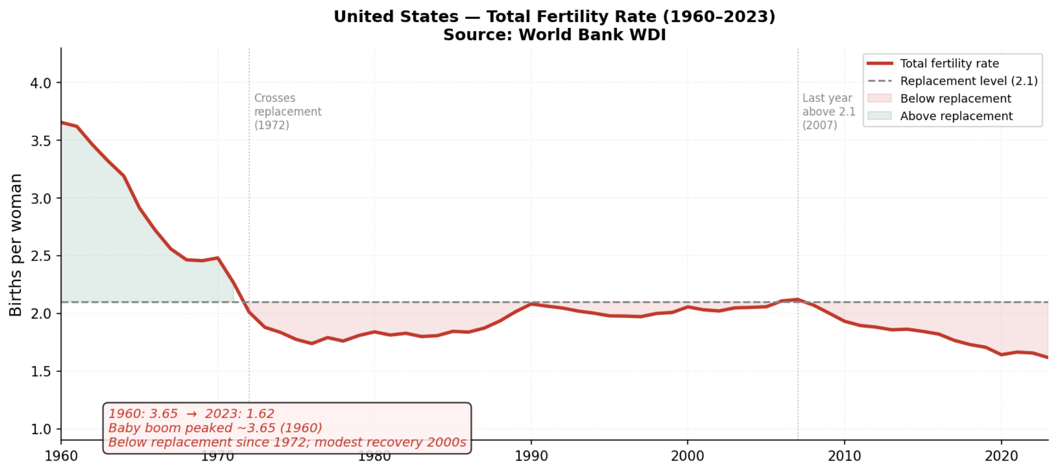

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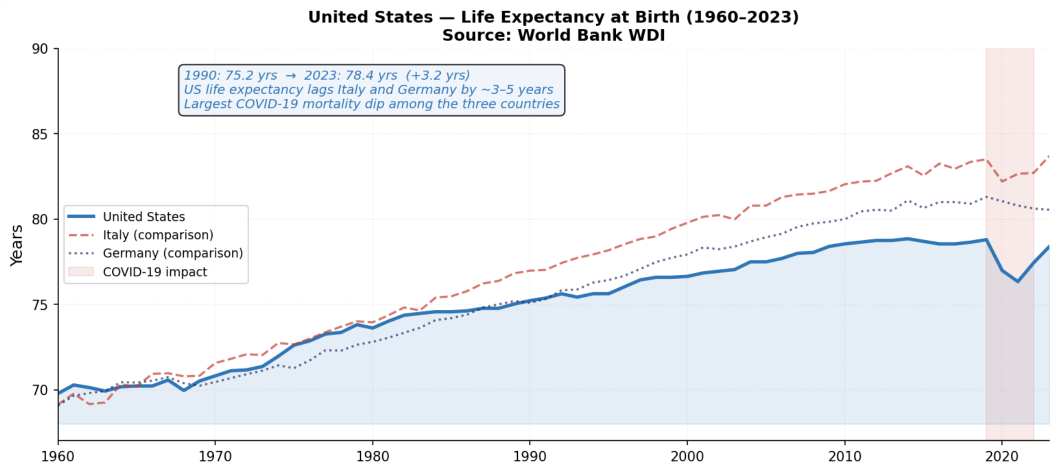

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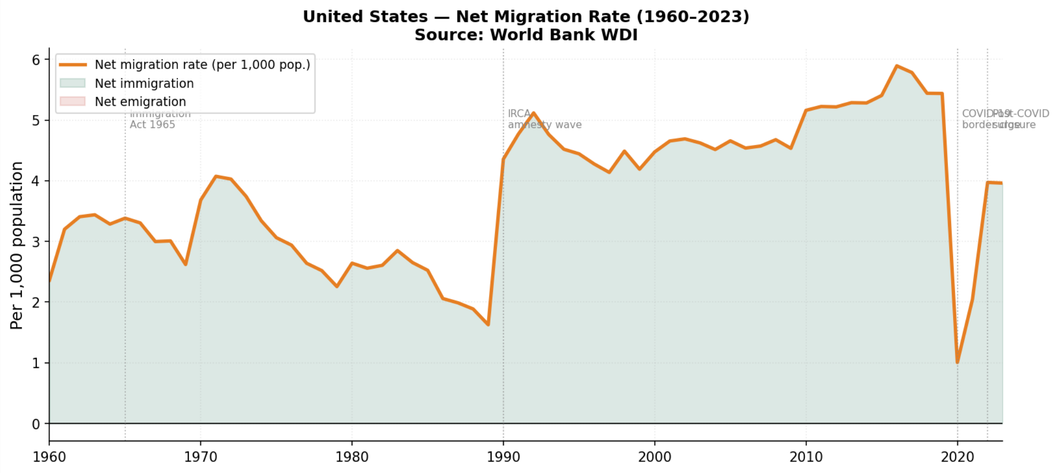

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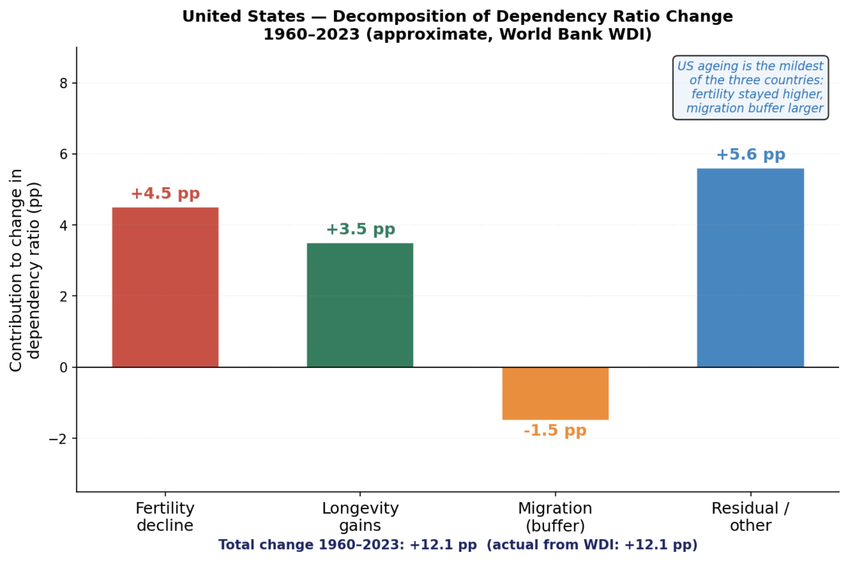
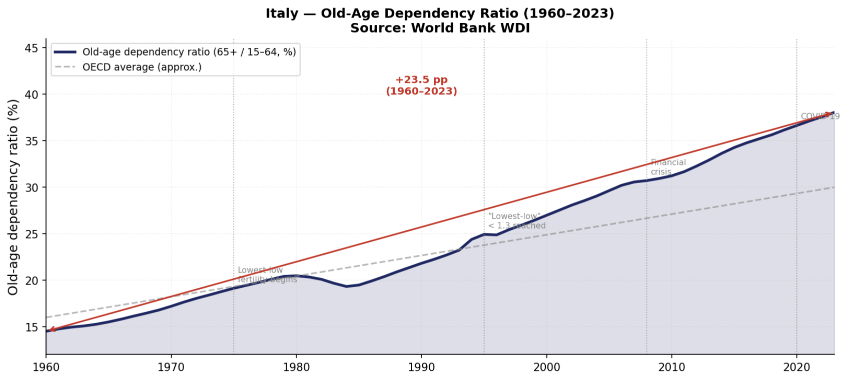


Italy Old-Age Dependency Ratio and its Determinants



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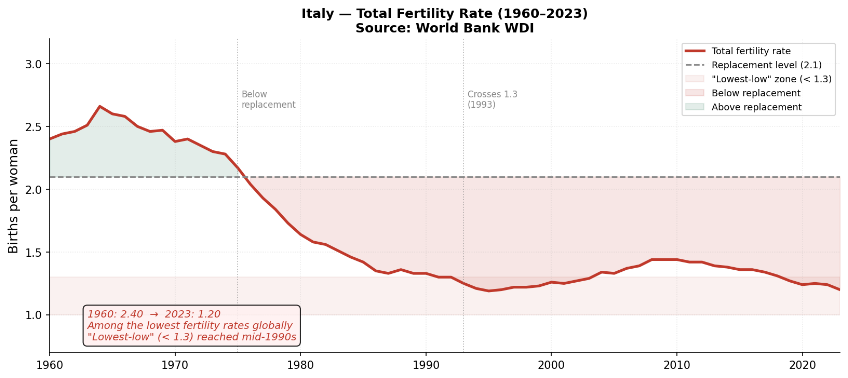

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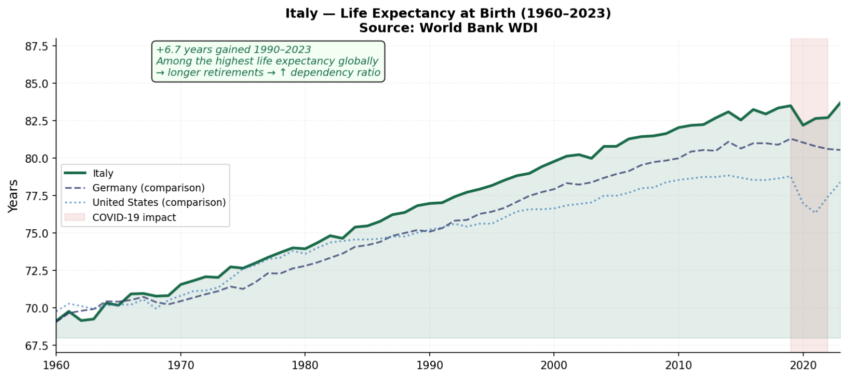

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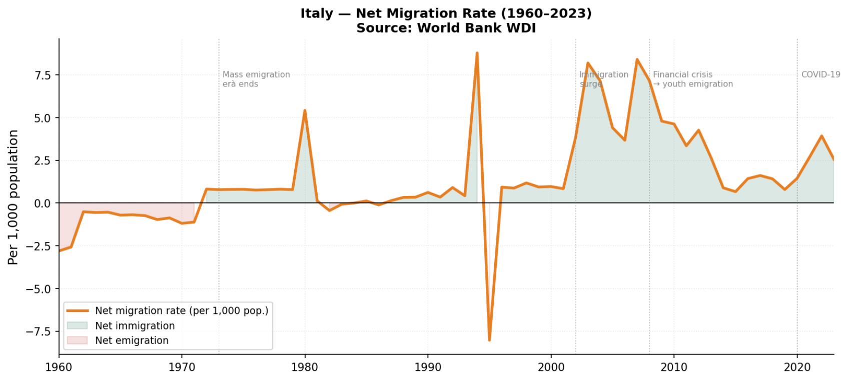

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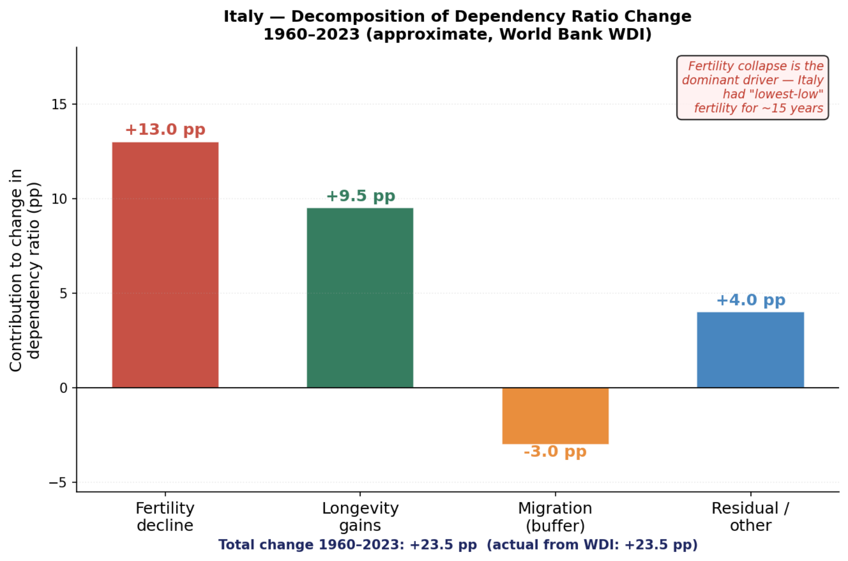
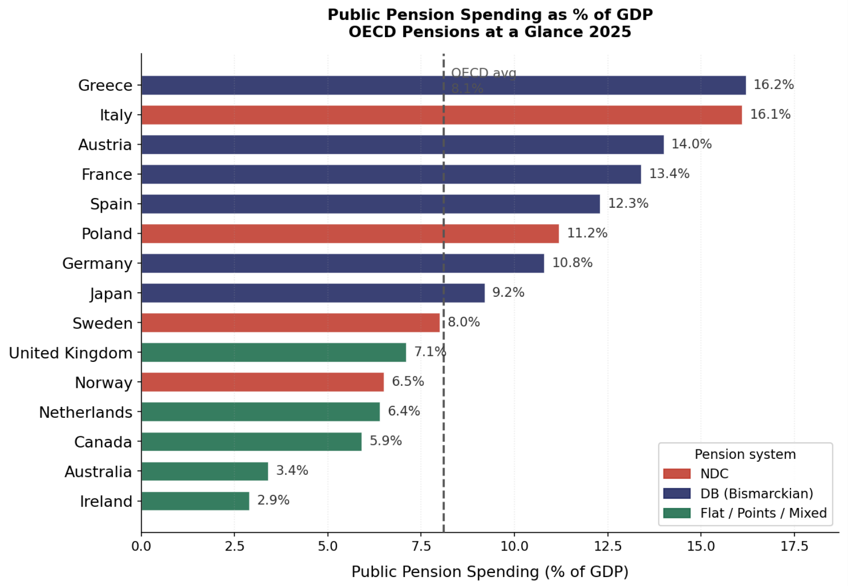


Pension Spending per GDP OECD Countries



Pension Payments and Contribution Rates

Message

If higher pension spending is financed only through payroll contributions, aging implies very large increases in contribution rates.

Fully Funded versus PAYG Public Pension Systems

Fully Funded Public Pension Systems i

- **Pay-as-you-go pensions**: current workers finance current retirees.
- **Fully funded pensions**: the government invests current contributions in capital markets and repays them during retirement.

Fully Funded Public Pension Systems ii

OLG model with fully funded pensions

$$w_t = c_t^1 + s_t + d_t, \tag{1a}$$
$$c_{t+1}^2 = (1 + r_{t+1})(s_t + d_t). \tag{1b}$$

- c_t^1 and c_{t+1}^2 denote young- and old-age consumption.
- s_t is private saving, d_t pension contributions, w_t the real wage, and r_t the real interest rate.
- Assume households and government receive the same return in capital markets.
- **Labor supply is inelastic** and normalized to one.

Fully Funded Public Pension Systems iii

Household lifetime utility is

$$U_t = u^y(c_t^1) + \beta u^o(c_{t+1}^2).$$

The Euler equation is

$$u_c^y(c_t^1) = \beta(1 + r_{t+1})u_c^o(c_{t+1}^2).$$

Equilibrium is characterized by

$$u_c^y\big(w_t - (s_t + d_t)\big) = \beta(1 + r_{t+1})u_c^o\big((1 + r_{t+1})(s_t + d_t)\big), \tag{2a}$$

$$s_t + d_t = (1 + n)k_{t+1}. \tag{2b}$$

Production satisfies

$$w_t = f(k_t) - k_t f_k(k_t), \qquad r_t = f_k(k_t).$$

Fully Funded Public Pension Systems iv

Neutrality result

Comparing $d_t = 0$ and $d_t > 0$, the allocation $\{c_t^1, c_{t+1}^2, k_{t+1}\}$ is unchanged. A fully funded pension mandate only relabels household asset holdings.

- Aggregate saving is unchanged.
- Capital accumulation is unchanged.
- Factor prices are unchanged.
- Household lifetime utility is unchanged.

Fully Funded Public Pension Systems v

Household lifetime utility with **elastic labor supply** is

$$U_t = u^y(c_t^1, l_t) + \beta u^o(c_{t+1}^2).$$

Fully funded contributions proportional to wage income

$$(1-\tau)w_t l_t = c_t^1 + s_t, \tag{3a}$$
$$c_{t+1}^2 = (1+r_{t+1})(d_t + s_t), \tag{3b}$$
$$d_t = \tau w_t l_t. \tag{3c}$$

The intertemporal budget constraint becomes

$$w_t l_t = c_t^1 + \frac{c_{t+1}^2}{1+r_{t+1}},$$

which does not depend on τ.

The household first-order conditions are

$$u_c^y(c_t^1, l_t) = \beta(1 + r_{t+1})u_c^o(c_{t+1}^2), \tag{4a}$$

$$-u_l^y(c_t^1, l_t) = w_t. \tag{4b}$$

The firm's conditions and capital-market equilibrium remain unchanged as well. Therefore even **a proportional wage tax is neutral in the fully funded benchmark**.

OLG Model with a PAYG Pension i

Pay-as-you-go variants:

1. benchmark with lump-sum contributions,
2. contributions proportional to wage income,
3. contributions-dependent pensions,
4. PAYG pensions with productivity growth.

Question

Once pensions are no longer funded, how do they affect aggregate saving, capital accumulation, and welfare?

OLG Model with a PAYG Pension ii

Benchmark PAYG model Social security is balanced every period:

$$N_t \tau_t = N_{t-1} pen_t \qquad \Rightarrow \qquad pen_t = (1+n) d_t.$$

Hence the implicit PAYG return is tied to population growth.

Household budget constraints are

$$w_t = c_t^1 + s_t + \tau_t, \tag{5a}$$

$$c_{t+1}^2 = (1 + r_{t+1}) s_t + pen_{t+1}. \tag{5b}$$

Therefore,

$$c_t^1 + \frac{c_{t+1}^2}{1 + r_{t+1}} = w_t - \tau_t + \frac{pen_{t+1}}{1 + r_{t+1}}.$$

OLG Model with a PAYG Pension iii

The Euler equation is unchanged in form,

$$u_c^y(c_t^1) = \beta(1 + r_{t+1})u_c^o(c_{t+1}^2),$$

but equilibrium now satisfies

$$u_c^y\left(w_t - s_t - \tau_t\right) = \beta(1 + r_{t+1})u_c^o\left((1 + r_{t+1})s_t + (1 + n)\tau_{t+1}\right).$$

For constant contributions, $\tau_{t+1} = \tau_t = \tau$, differentiation gives

$$\frac{\partial s_t}{\partial \tau} = -\frac{u_{cc}^y + \beta(1 + n)(1 + r_{t+1})u_{cc}^o}{u_{cc}^y + \beta(1 + r_{t+1})^2 u_{cc}^o} < 0.$$

Thus higher PAYG contributions reduce private saving.

OLG Model with a PAYG Pension iv

In general equilibrium,

$$(1+n)k_{t+1} = s\big(w(k_t), r(k_{t+1}), \tau\big).$$

Differentiating with respect to τ yields

$$\frac{dk_{t+1}}{d\tau} = \frac{\partial s_t / \partial \tau_t}{1 + n - s_r f_{kk}(k_{t+1})} < 0$$

whenever $s_r > 0$ and $f_{kk} < 0$.

Conclusion

A larger PAYG system crowds out capital in the benchmark OLG economy.

OLG Model with a PAYG Pension v

Numerical example with inelastic labor, run Social_security_baseline.ipynb

Lifetime utility is

$$U_t = \ln c_t^1 + \beta \ln c_{t+1}^2 - \nu_0 \frac{l_t^{1+\nu_1}}{1+\nu_1},$$

with fixed labor $\bar{l} = 0.3$ and constant pension contribution $\tau_t = \tau$. The intertemporal budget constraint becomes

$$w_t \bar{l} + \frac{n - r_{t+1}}{1 + r_{t+1}} \tau = c_t^1 + \frac{c_{t+1}^2}{1 + r_{t+1}}.$$

First-order conditions imply

$$c_{t+1}^2 = \beta(1 + r_{t+1}) c_t^1.$$

Substituting into the intertemporal budget constraint gives

$$c_t^1 = \frac{1}{1+\beta}\left[w_t\bar{l} + \frac{n - r_{t+1}}{1 + r_{t+1}}\tau\right],$$

so private saving is

$$s_t = \frac{\beta}{1+\beta} w_t\bar{l} - \frac{1+\beta+\beta r_{t+1}+n}{(1+\beta)(1+r_{t+1})}\tau.$$

Capital market equilibrium $s_t = (1+n)k_{t+1}$ then yields the law of motion

$$(1+n)k_{t+1} = \frac{\beta}{1+\beta} w_t\bar{l} - \frac{1+\beta+\beta r_{t+1}+n}{(1+\beta)(1+r_{t+1})}\tau.$$

OLG Model with a PAYG Pension vii

With Cobb-Douglas production,

$$Y_t = K_t^{\alpha}(N_t l_t)^{1-\alpha},$$

factor prices satisfy

$$w_t = (1-\alpha)k_t^{\alpha} l_t^{-\alpha}, \tag{6a}$$
$$r_t = \alpha k_t^{\alpha-1} l_t^{1-\alpha}. \tag{6b}$$

Calibration using:

$$\alpha = 0.36, \qquad \beta = 0.40 \to \beta_{\text{annual}} \approx 0.967, \qquad n = 0.10, \qquad \bar{l} = 0.30,$$

and compare $\tau = 0$ against a replacement rate of 30%.

OLG Model with a PAYG Pension viii

Consumption-equivalent change If old steady-state utility is

$$U = \ln c^1 + \beta \ln c^2 - \nu_0 \frac{\bar{l}^{1+\nu_1}}{1+\nu_1}$$

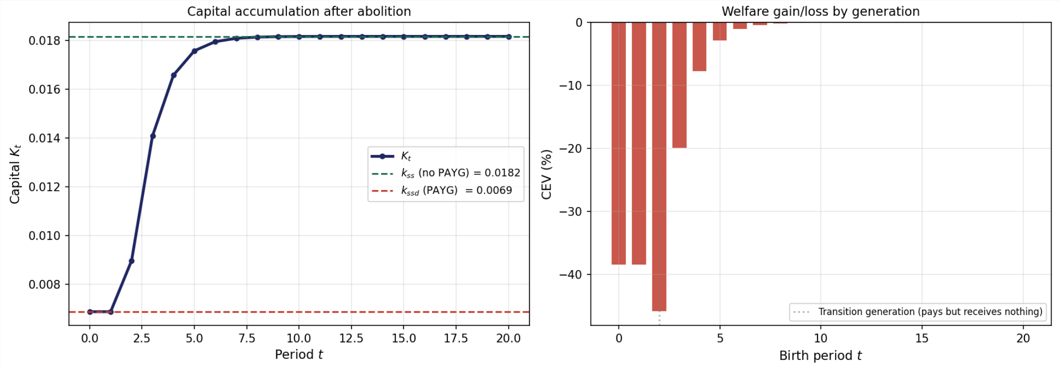
and the new steady-state utility is $\tilde{U}$, then Δ is defined by

$$\tilde{U} = \ln\big((1+\Delta)c^1\big) + \beta \ln\big((1+\Delta)c^2\big) - \nu_0 \frac{\bar{l}^{1+\nu_1}}{1+\nu_1}.$$

With logarithmic utility and unchanged labor,

$$(1+\beta)\ln(1+\Delta) = \tilde{U} - U.$$

Dynamics of the Capital Stock After Abolishing PAYG Pensions



PAYG Pensions with Elastic Labor Supply

PAYG Pensions with Elastic Labor Supply i

Contributions are levied on labor income:

$$d_t = \tau w_t l_t.$$

Household budget constraints are

$$\begin{aligned}(1-\tau)w_t l_t &= c_t^1 + s_t, \\ c_{t+1}^2 &= (1+r_{t+1})s_t + (1+n)d_{t+1}.\end{aligned}$$

Hence,

$$(1-\tau)w_t l_t + \frac{1+n}{1+r_{t+1}} d_{t+1} = c_t^1 + \frac{c_{t+1}^2}{1+r_{t+1}}.$$

PAYG Pensions with Elastic Labor Supply ii

Maximization implies

$$\frac{1}{c_t^1} = \lambda_t, \tag{7a}$$

$$\frac{\beta}{c_{t+1}^2} = \frac{\lambda_t}{1 + r_{t+1}}, \tag{7b}$$

$$\nu_0 l_t^{\nu_1} = \lambda_t (1 - \tau) w_t. \tag{7c}$$

The Frisch elasticity is $1/\nu_1$.

PAYG Pensions with Elastic Labor Supply iii

Combining these conditions gives

$$c_t^1 = \frac{1}{1+\beta}\left[(1-\tau)w_t l_t + \frac{1+n}{1+r_{t+1}}d_{t+1}\right],$$

and savings

$$s_t = \frac{\beta}{1+\beta}(1-\tau)w_t l_t - \frac{1}{1+\beta}\frac{1+n}{1+r_{t+1}}d_{t+1}.$$

In equilibrium,

$$(1+n)k_{t+1} = \frac{\beta}{1+\beta}(1-\tau)w_t l_t - \frac{1}{1+\beta}\frac{1+n}{1+r_{t+1}}d_{t+1}.$$

PAYG Pensions with Elastic Labor Supply iv

Run Social_security_elastic_ipynb

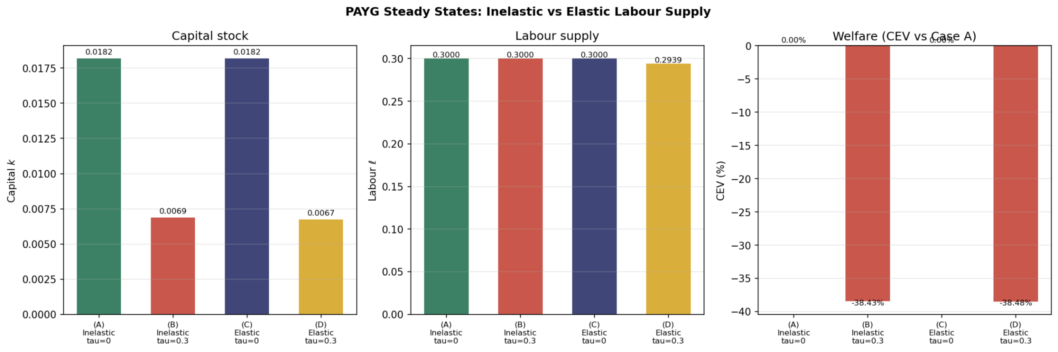
- Calibration in the source sets $\nu_1 = 3.33$ and $\nu_0 = 257.15$ to match $l = 0.3$.
- With endogenous labor, the output and welfare losses from PAYG are even larger than in the inelastic-labor benchmark.
- In the local source, labor falls by about 2% and steady-state output falls by about 30.8%.

PAYG Steady-State Comparison

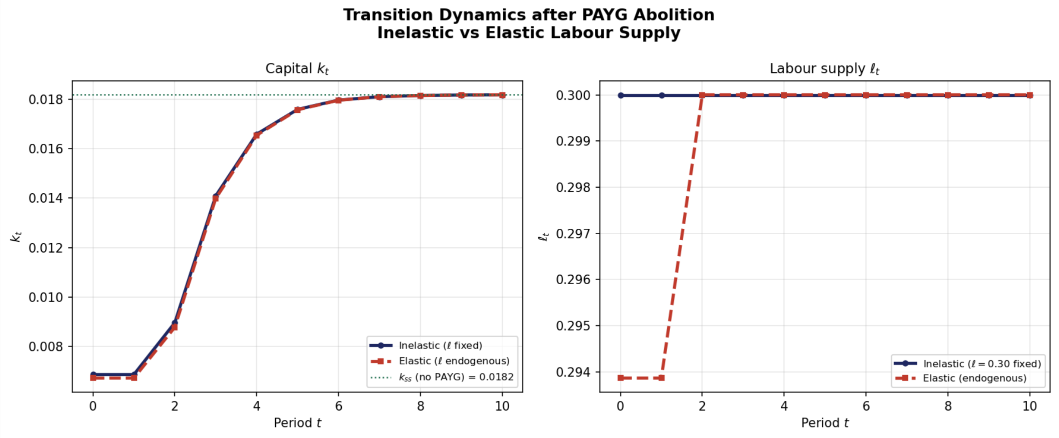
Table 1: PAYG Steady-State Comparison

	(A) Inelastic, $\tau = 0$ [reference]	(B) Inelastic, $\tau = 0.3$	(C) Elastic, $\tau = 0.3$
Capital k	0.0182	0.0069	0.0067
Labour ℓ	0.3000	0.3000	0.2939
Output y	0.1093	0.0770	0.0754
Wage w	0.2333	0.1643	0.1643
Interest rate r (%)	2.1656	4.0384	4.0384
Consumption c_1	0.0500	0.0269	0.0264
Consumption c_2	0.0633	0.0543	0.0532
Lifetime utility U	-4.4234	-5.1023	-5.1036
CEV vs. Case (A) (%)	[0.00]	-38.43	-38.48

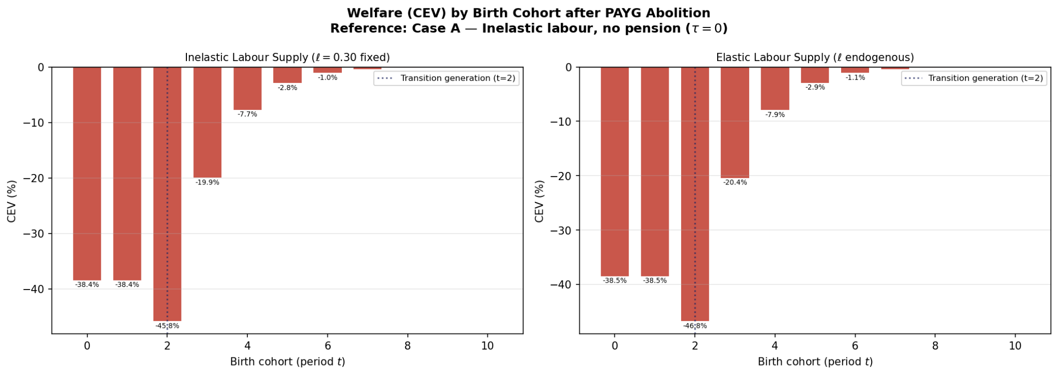
PAYG Steady-State Comparison



PAYG Transition Comparison



PAYG Transition Comparison



Transition Effects of Pension Reform

- The capital stock converges only gradually to the new steady state.
- Even when long-run welfare rises, the transition generation can lose because it still finances current retirees but receives much lower benefits later on.
- The first reform generation experiences a large negative consumption-equivalent change relative to the new steady state.

Policy implication

Pension reform is not just a steady-state question. It is a transition-design and cohort-compensation problem.

PAYG Pensions with Contribution-Based Benefits

PAYG with Contribution-Based Benefits

Let contributions be

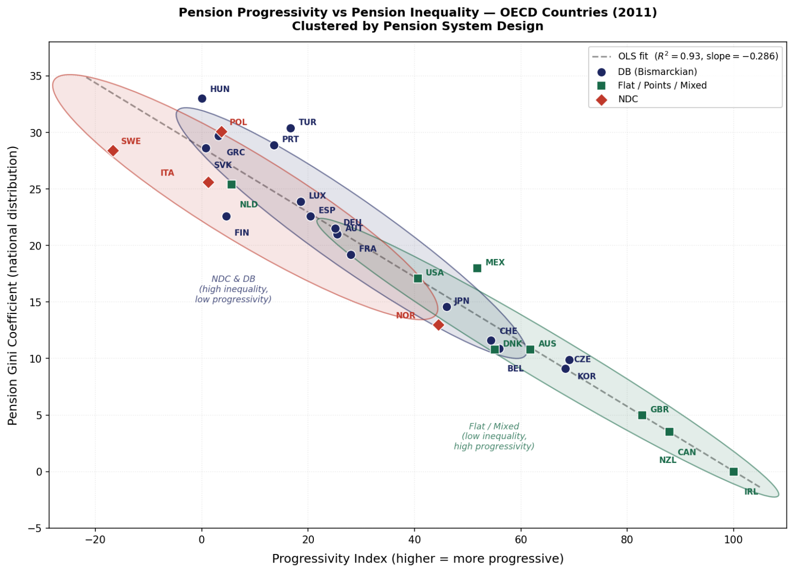
$$x_t = \tau w_t l_t,$$

and pension benefits follow

$$pen_{t+1} = pen_{\min} + \theta x_t.$$

- $\theta = 0$: pure flat-rate pensions.
- $pen_{\min} = 0$: pensions fully proportional to pre-retirement income.
- Higher $pen_{\min}$ means a more progressive system.

PAYG with Contribution-Based Benefits



PAYG with Contribution-Based Benefits

- Budget constraint:

$$(1-\tau)\, w_t\, l_t + \frac{pen_{t+1}}{1+r_{t+1}} = c_t^1 + \frac{c_{t+1}^2}{1+r_{t+1}}$$

- First-order conditions:

$$\frac{1}{c_t^1} = \lambda_t, \tag{8a}$$

$$\frac{\beta}{c_{t+1}^2} = \frac{\lambda_t}{1+r_{t+1}}, \tag{8b}$$

$$\nu_0\, l_t^{\nu_1} = \lambda_t (1-\tau)\, w_t + \lambda_t\, \frac{\theta\, \tau\, w_t}{1+r_{t+1}}. \tag{8c}$$

- PAYG budget balance:

$$N_t\, x_t = N_{t-1}\, pen_t$$

PAYG with Contribution-Based Benefits

The labor first-order condition becomes

$$\nu_0 l_t^{\nu_1} = \lambda_t(1-\tau)w_t + \lambda_t \frac{\theta\tau w_t}{1+r_{t+1}},$$

which contains an additional term from the earnings-related pension link. This partially offsets the labor wedge from payroll taxation.

Result

Contributions-defined pensions are an intermediate case between pure lump-sum PAYG and pure wage-tax financing: labor declines less, but aggregate welfare still falls substantially at high replacement rates.

PAYG with Contribution-Based Benefits

In the stationary state:

$$c^2 = c^1 \beta (1 + r), \tag{9a}$$

$$\tau w l = \frac{pen_{\min} + \theta \tau w l}{1 + r}, \tag{9b}$$

$$w = (1 - \alpha) k^\alpha l^{-\alpha}, \tag{9c}$$

$$r = \alpha k^{\alpha - 1} l^{1-\alpha}, \tag{9d}$$

$$\nu_0 l^{\nu_1} c^1 = (1 - \tau) w + \frac{\theta \tau w}{1 + r}, \tag{9e}$$

$$c^1 + \frac{c^2}{1 + r} = (1 - \tau) w l + \frac{pen_{\min} + \theta \tau w l}{1 + r}, \tag{9f}$$

$$(1 + n) k = (1 - \tau) w l - c^1. \tag{9g}$$

PAYG with Contribution-Based Benefits

Starting from the PAYG budget balance:

$$N_t \cdot \tau w \ell = N_{t-1} \left(pen_{\text{min}} + \theta \, \tau w \ell \right)$$

Dividing by N_{t-1} and using $N_t / N_{t-1} = 1 + n$:

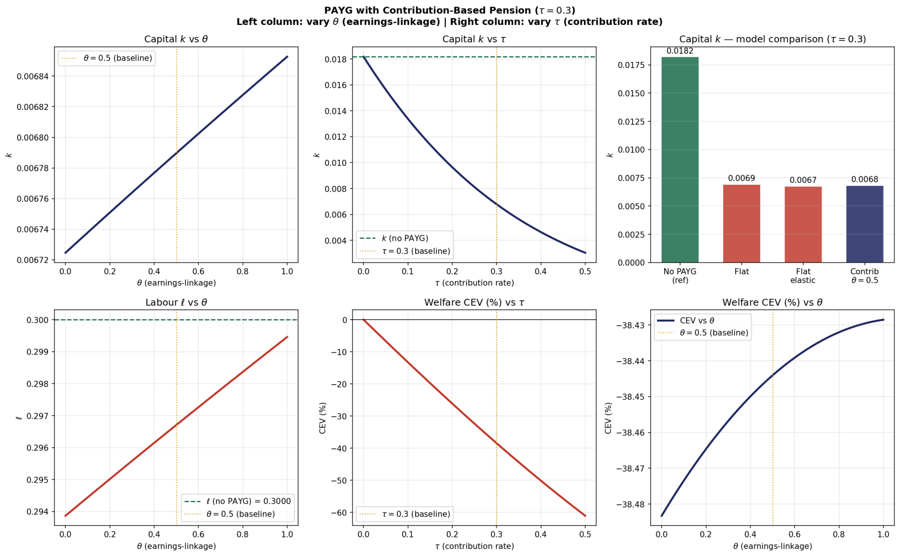
$$(1+n) \, \tau w \ell = pen_{\text{min}} + \theta \, \tau w \ell$$

Solving for pen_{min}:

$$pen_{\text{min}} = \tau w \ell \, (1 + n - \theta)$$

Policy experiments: Check how CE is affected by changes in progressivity θ and contribution rate τ. **Run Social_security_progressivity.ipynb**

Policy Experiments



Contribution-Based PAYG: Effect of Earnings-Linkage θ

- As θ rises from 0 to 1 (fixing $\tau = 0.3$):
    - **Capital** k **rises** toward the no-PAYG level: part of the contribution returns as pension, reducing the implicit tax on saving.
    - **Labour** ℓ **rises** toward the no-PAYG level: the effective labour tax falls because working more raises the future pension — the labour FOC gains the additional term $\frac{\theta\,\tau\,w}{1+r}$.
    - **Welfare CEV rises** (less negative): PAYG becomes less distortionary. At a threshold θ^* the CEV crosses zero — beyond this the actuarial linkage more than compensates for the low implicit return $n < r$.

Contribution-Based PAYG: Effect of Contribution Rate τ

- As τ rises from 0 to 0.5 (fixing $\theta = 0.5$):
  - **Capital falls** monotonically: higher contributions crowd out more private saving.
  - **Welfare CEV falls** monotonically: a higher contribution rate amplifies every distortion.
- **Cross-model comparison** at $\tau = 0.3$:

Model	Capital loss	Welfare cost (CEV)
Flat pension, inelastic ℓ	-62%	-38.4%
Flat pension, elastic ℓ	-63%	-38.5%
Contrib.-based $\theta = 0.5$, elastic ℓ	Smaller	Less negative

$\rightarrow$ The contribution-based design makes PAYG behave more like a forced savings plan and less like a pure payroll tax — reducing both the capital and labour distortions.

PAYG Pensions with Productivity Growth

PAYG Pension and Productivity Growth i

Public Pensions and Growth

- Production: $Y_t = K_t^{\alpha}\left(A_t N_t l_t\right)^{1-\alpha}$.

- Labor-augmenting technology A_t grows at the exogenous rate g:

$$A_t = (1+g)A_{t-1}.$$

- We set $g = 0.80 = 80\%$ so that output per capita grows at approximately 2.0% per year (periods correspond to 30 years).

PAYG Pension and Productivity Growth ii

- Budget constraint:

$$s_t = (1-\tau)w_t A_t l_t - c_t^1, \tag{10a}$$
$$c_{t+1}^2 = (1+r_{t+1})s_t + d_{t+1}, \tag{10b}$$
$$c_t^1 + \frac{c_{t+1}^2}{1+r_{t+1}} = (1-\tau)w_t A_t l_t + \frac{d_{t+1}}{1+r_{t+1}}. \tag{10c}$$

- First-order conditions:

$$c_{t+1}^2 = \beta(1+r_{t+1})c_t^1, \tag{11a}$$
$$\nu_0 l_t^{\nu_1} = \frac{(1-\tau)w_t A_t}{c_t^1}, \tag{11b}$$

PAYG Pension and Productivity Growth iii

and in stationary values $\tilde{c}_t^1 \equiv c_t^1/A_t$, $\tilde{c}_{t+1}^2 \equiv c_{t+1}^2/A_{t+1}$:

$$(1+g)\tilde{c}_{t+1}^2 = \beta(1+r_{t+1})\tilde{c}_t^1, \tag{12a}$$

$$\nu_0 l_t^{\nu_1} = \frac{(1-\tau)w_t}{\tilde{c}_t^1}, \tag{12b}$$

implying equilibrium savings in stationary values, $\tilde{s}_t \equiv s_t/A_t$:

$$\tilde{s}_t = \frac{\beta}{1+\beta} w_t l_t - \frac{1}{1+\beta} \frac{(1+g)\tilde{d}_{t+1}}{1+r_{t+1}}.$$

- In equilibrium, the social security budget is balanced

$$N_t d_{t+1} = N_{t+1} \tau w_{t+1} A_{t+1} l_{t+1},$$

and, with $\tilde{d}_t \equiv d_t/A_t$, $\tilde{d}_{t+1} = (1+n)\tau w_{t+1} l_{t+1}$.

PAYG Pension and Productivity Growth iv

- Firms maximize profits:

$$\Pi_t = K_t^{\alpha}\left(A_t N_t l_t\right)^{1-\alpha} - w_t A_t N_t l_t - r_t K_t,$$

implying the first-order conditions

$$w_t = (1-\alpha)\left(\frac{K_t}{A_t N_t}\right)^{\alpha} l_t^{-\alpha} = k_t^{\alpha} l_t^{-\alpha}, \tag{13a}$$

$$r_t = \alpha k_t^{\alpha-1} l_t^{1-\alpha}, \tag{13b}$$

with $k \equiv K/(AN)$.

- In capital market equilibrium,

$$N_t s_t = K_{t+1},$$

or, after dividing by $A_t N_t$ on both sides $\tilde{s}_t = k_{t+1}(1+g)(1+n) = \frac{\beta}{1+\beta} w_t l_t - \frac{1}{1+\beta}\frac{(1+g)\tilde{d}_{t+1}}{1+r_{t+1}}$.

PAYG Pension and Productivity Growth v

- Steady state:

$$(1+g)\tilde{c}^2 = \beta(1+r)\tilde{c}^1, \tag{14a}$$

$$\nu_0 l^{\nu_1}\tilde{c}^1 = (1-\tau)w, \tag{14b}$$

$$w = k^{\alpha} l^{-\alpha}, \tag{14c}$$

$$r = \alpha k^{\alpha-1} l^{1-\alpha}, \tag{14d}$$

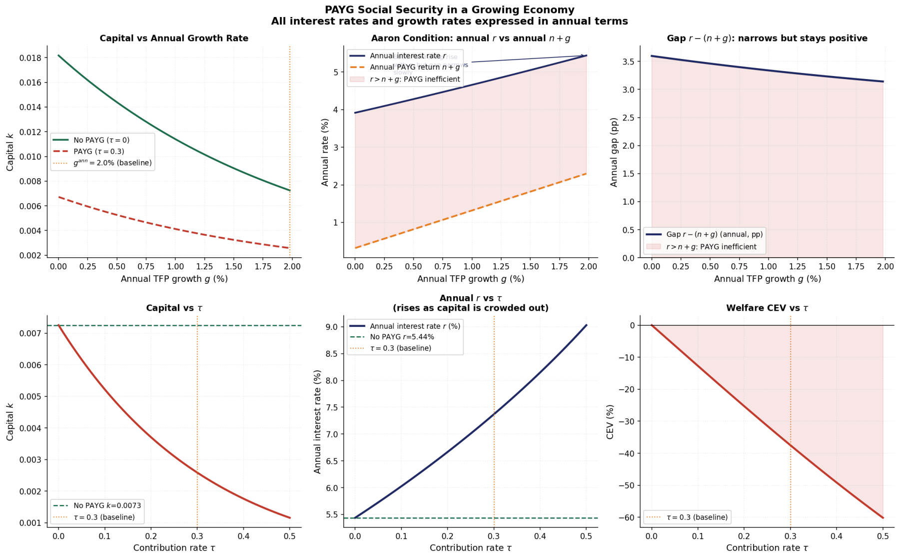
$$k(1+g)(1+n) = \frac{\beta}{1+\beta}wl - \frac{1}{1+\beta}\frac{(1+g)\tilde{d}}{1+r}, \tag{14e}$$

$$\tilde{d} = (1+n)\tau wl, \tag{14f}$$

$$\tilde{c}^1 + \frac{(1+g)\tilde{c}^2}{1+r} = (1-\tau)wl + \frac{(1+g)\tilde{d}}{1+r}. \tag{14g}$$

Run Social_security_growth.ipynb

PAYG Pension and Productivity Growth



PAYG and Growth: The Aaron Condition

Key result

Both the market return r and the PAYG return $n + g$ rise as g increases — but $n + g$ rises marginally faster, so the gap narrows.

Why does r rise with g? The mechanism runs through the capital market:

1. Higher g shifts the growth-adjusted Euler equation

$$c_1 \, \beta_0 \, (1 + r) = (1 + g) \, c_2$$

   Households need to accumulate *less* wealth today because future consumption grows automatically.

2. Lower saving reduces the equilibrium capital stock k.

3. Lower k raises the marginal product of capital:

$$r = \alpha \, k^{\alpha - 1} \, \ell^{1-\alpha} \quad \text{is decreasing in } k.$$

PAYG and Growth: Numerical Evidence

Annual rates (model period = 30 years, converted via $(1+x)^{1/30}-1$):

Annual g	Annual r	Annual $n+g$	Gap $r-(n+g)$
0.00%	3.92%	0.32%	3.60 pp
1.00%	4.76%	1.26%	3.50 pp
1.98%	5.44%	2.30%	3.14 pp

- The gap narrows from 3.60 pp to 3.14 pp as g rises.
- The Aaron condition $n+g>r$ is **never satisfied** in this calibration.
- The market return *always* exceeds the PAYG implicit return $\Rightarrow$ PAYG remains **dynamically inefficient** for all values of g considered.

PAYG and Growth: Effect of Contribution Rate τ

Fixing $g = g^{\text{target}}$ and raising τ from 0 to 0.5:

- **Capital** k **falls** monotonically: higher τ crowds out private saving.
- **Annual interest rate** r **rises**: lower k raises the marginal product of capital.
- **Welfare CEV falls** monotonically: higher τ amplifies every distortion — both the saving wedge and the labour wedge.

PAYG and Growth: Bottom Line

What growth does to PAYG

Higher g makes PAYG *marginally* less distortionary by raising $n+g$ slightly faster than r, narrowing the gap from 3.60 pp to 3.14 pp (annually).

What growth does not do

Growth does not make PAYG efficient. In any plausible calibration for developed economies:

$$r \approx 4\% \quad > \quad n+g \approx 2\%$$

The Aaron condition $n+g>r$ is not satisfied.

When is PAYG Pareto-improving?

Only in a **dynamically inefficient** economy — one where capital has been over-accumulated to the point that $r<n+g$.

Summary

1. Fully funded pension mandates are neutral in the frictionless OLG benchmark.
2. PAYG pensions reduce private saving and crowd out capital.
3. With elastic labor, payroll financing adds an extra distortion and deepens steady-state losses.
4. Aging makes generous pension systems harder to sustain and narrows fiscal space.
5. Transition dynamics are central because many currently alive cohorts can lose from otherwise beneficial reforms.